

The Evolution of Language and Themes in *The Hunger Games* Trilogy

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Abstract

This project analyzes linguistic, emotional, and thematic changes throughout Suzanne Collins' *The Hunger Games* trilogy using computational text mining methods in R. The novels were divided into 81 chapter-level documents and analyzed using word frequency, TF-IDF, sentiment analysis, and topic modeling. Statistical methods including ANOVA, Chi-square tests, and regression were used to evaluate differences between books. Our results reveal a clear progression in tone and language from survival-focused themes in *The Hunger Games*, to political tension in *Catching Fire*, and finally to rebellion and warfare in *Mockingjay*.

1 Introduction

Suzanne Collins' *The Hunger Games* trilogy has sold over 100 million copies and its film adaptations have grossed over three billion dollars globally. Clearly, this series has made a large impact worldwide and has themes surrounding politics and power that are still very relevant today. The series follows protagonist Katniss Everdeen through a progression of survival, political upheaval, and revolution. Beyond its narrative significance, the trilogy provides a good case for computational text analysis. This project examines the character, thematic, and linguistic development both within each novel and across the trilogy as a whole. The study analyzes sentiment, vocabulary, themes, and shifts in character sentiment throughout the texts. Using statistical text mining and hypothesis-testing methods in R, the project investigates how language and narrative patterns evolve across the trilogy.

While there are various research papers and analyses on this trilogy, most prior studies focus on themes such as authoritarian critique, gender roles, and the psychological development of the characters using traditional literary analysis methods. Comparatively little work has applied computational text mining techniques to examine how linguistic patterns, sentiment, and thematic emphasis evolve across the series. This project addresses that gap by applying text analysis methods taught in this course to the trilogy. By using statistical and computational tools, this study explores whether these approaches reveal patterns consistent with conclusions drawn from traditional literary analysis or provide additional insights into the narrative progression of the series.

1.1 Research Questions

This project investigates the following questions:

1. How does sentiment shift across the three novels and does it become more negative as the narrative progresses?
2. How do word frequency patterns change from novel to novel, particularly for words related to survival, violence, power, and rebellion?
3. Does the usage of certain words increase as the narrative evolves from individual survival to political revolution?
4. How does Suzanne Collins' writing style differ between the three books?

2 Dataset and Study Design

2.1 Data Collection

The data set consists of the three novels in the trilogy:

- *The Hunger Games*
- *Catching Fire*
- *Mockingjay*

Digital copies of the novels were obtained online in PDF format and imported into R for analysis. Each book was divided into chapters, resulting in a dataset of 81 documents (27 chapters per book). Chapters were used as the unit of analysis to increase sample size and therefore the statistical power of our results.

2.2 Preprocessing

Standard text preprocessing steps were applied using the `tidytext` framework in R:

- Tokenization
- Lowercase conversion
- Removed:
 - Punctuation
 - Non-letter characters
 - English stop words
 - Empty rows & single-letter words
 - Custom stop words
 - Sparse terms

The resulting corpus was used to compute word frequencies, sentiment scores, TF-IDF values, and topic models.

3 Exploratory Text Analysis

3.1 Book 1: *The Hunger Games*

3.1.1 Sentiment Over Time

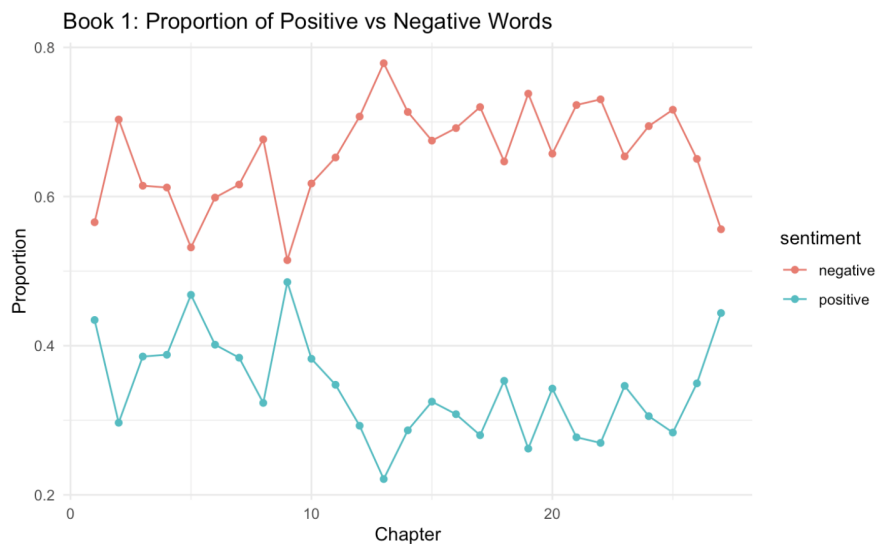


Figure 1: Proportion of Positive vs Negative Words in Book 1

Using the Bing sentiment lexicon, the graph above shows that negative sentiment consistently outweighs positive sentiment across Book 1. Early chapters are characterized by greater emotional variance, followed by a gradual convergence of sentiment towards the end of the book.

3.1.2 NRC Sentiment

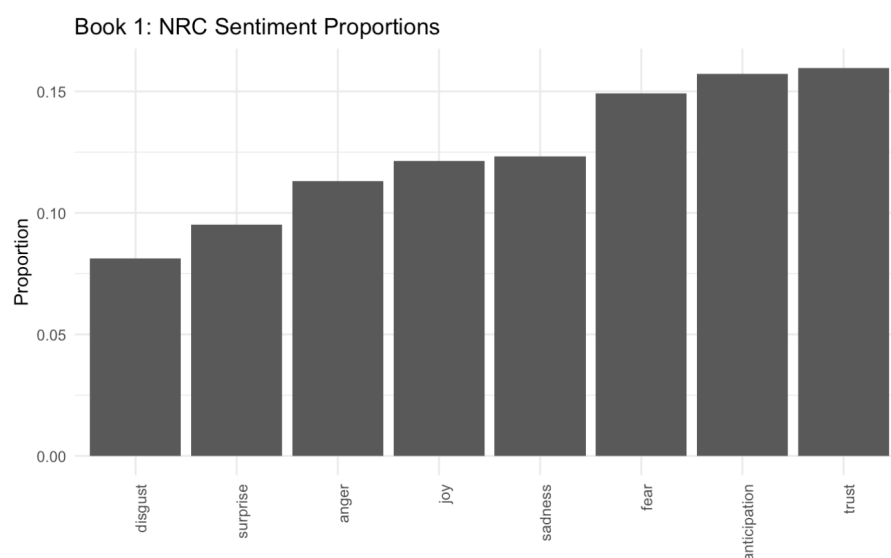


Figure 2: NRC Sentiment Analysis Book 1

The NRC sentiment distribution reveals that trust, anticipation, and fear make up the greatest proportion (approx. 15%-16% each) of emotional prevalence in Book 1 while disgust represents the least prominent sentiment.

3.1.3 Vocabulary

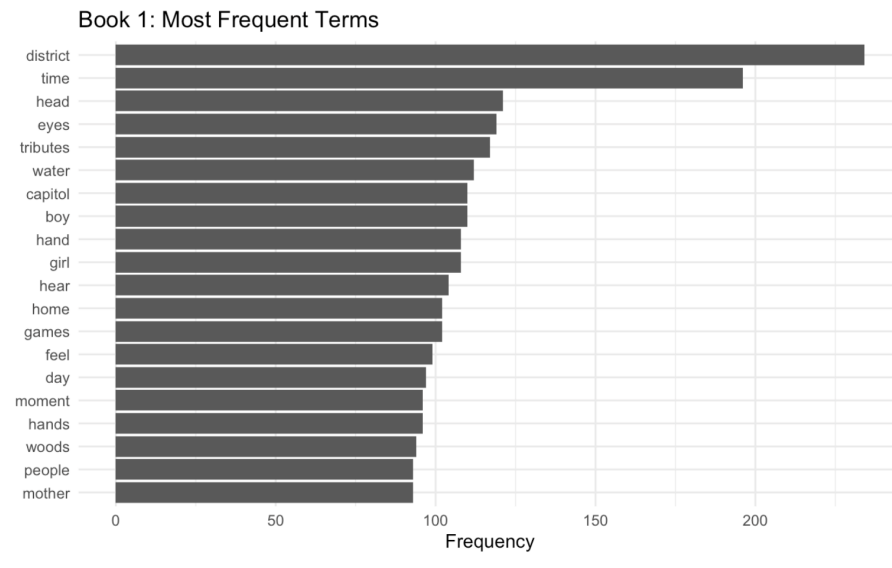


Figure 3: 20 Most Frequent Terms in Book 1

The horizontal bar chart illustrates the most frequent terms in Book 1, highlighting key narrative motifs such as 'district,' 'time,' and 'tributes.' These high-frequency words reflect the novel's primary focus on the survival stakes of the Games.

3.1.4 Themes Detected Via Topic Modeling

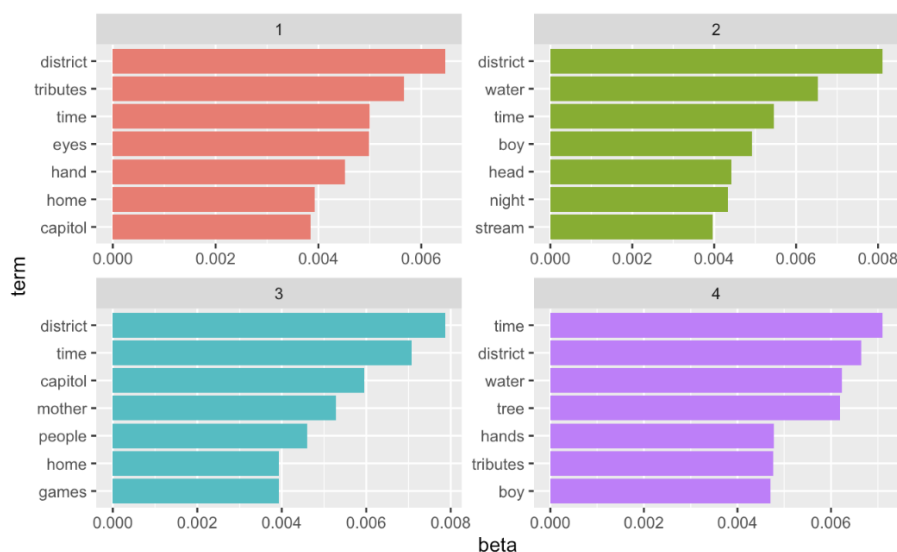


Figure 4: Themes Detected via LDA Topic Modeling in Book 1

The figure above displays the themes detected via LDA topic modeling. We categorized these as: The Games & Physical Experience (Topic 1), Survival in the Arena (Topic 2), Home & Society (Topic 3), and The Arena & Combat (Topic 4).

3.2 Book 2: *Catching Fire*

3.2.1 Sentiment Over Time

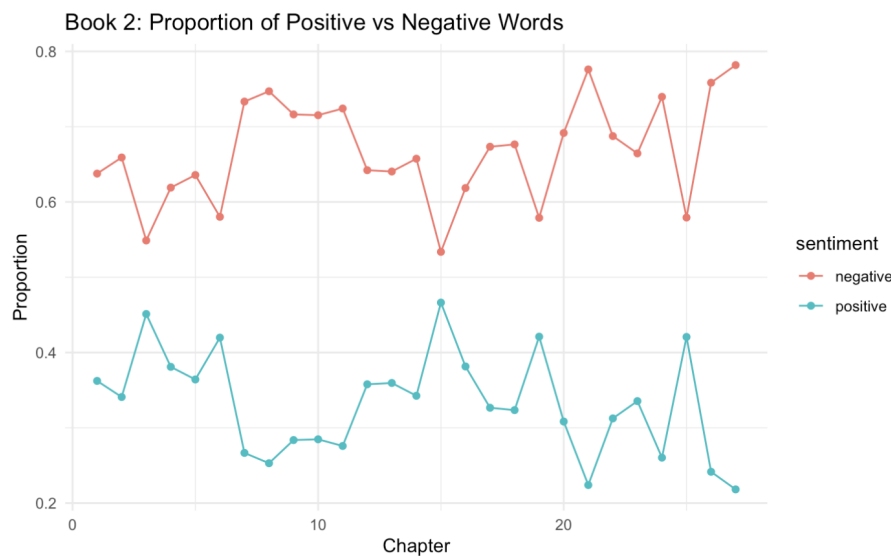


Figure 5: Proportion of Positive vs Negative Words in Book 2

The sentiment analysis for *Catching Fire* (Book 2) reveals a persistent dominance of negative sentiment over positive sentiment. In comparison to the first book, Book 2 exhibits higher emotional volatility, with the gap between negative and positive sentiment diverging the most towards the end of the book.

3.2.2 NRC Sentiment

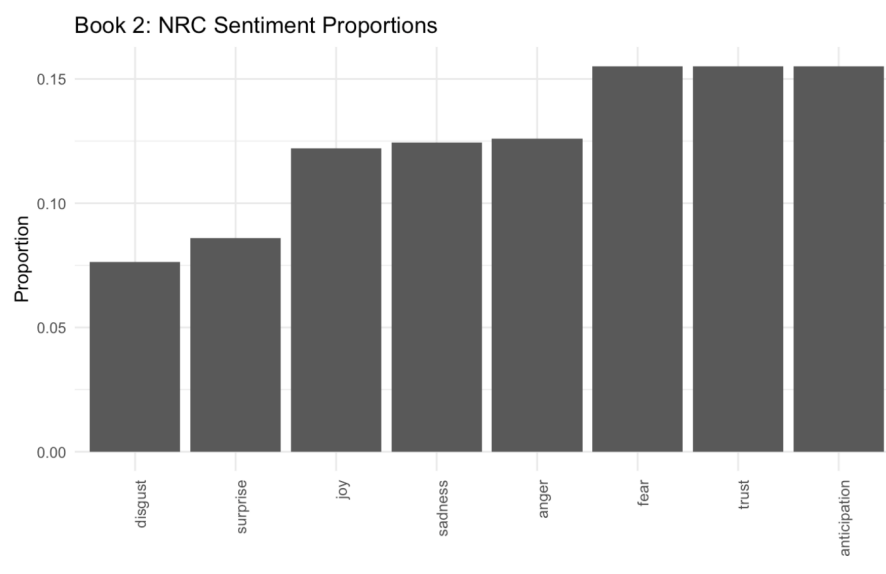


Figure 6: NRC Sentiment Analysis Book 2

Similar to book 1, the NRC sentiment distribution for Book 2 shows that trust, anticipation, and fear make up the greatest proportion (approx. 16% each) of emotional prevalence while disgust makes up the smallest proportion.

3.2.3 Vocabulary

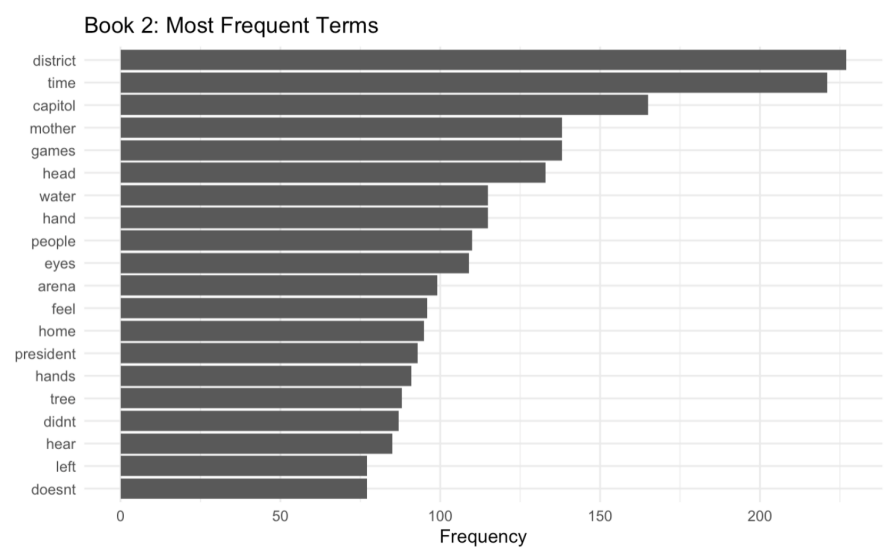


Figure 7: 20 Most Frequent Terms in Book 2

The horizontal bar chart for Book 2 illustrates a significant lexical overlap with Book 1. Recurrent terms such as 'district,' 'time,' and 'water' persist as dominant features.

3.2.4 Themes Detected Via Topic Modeling

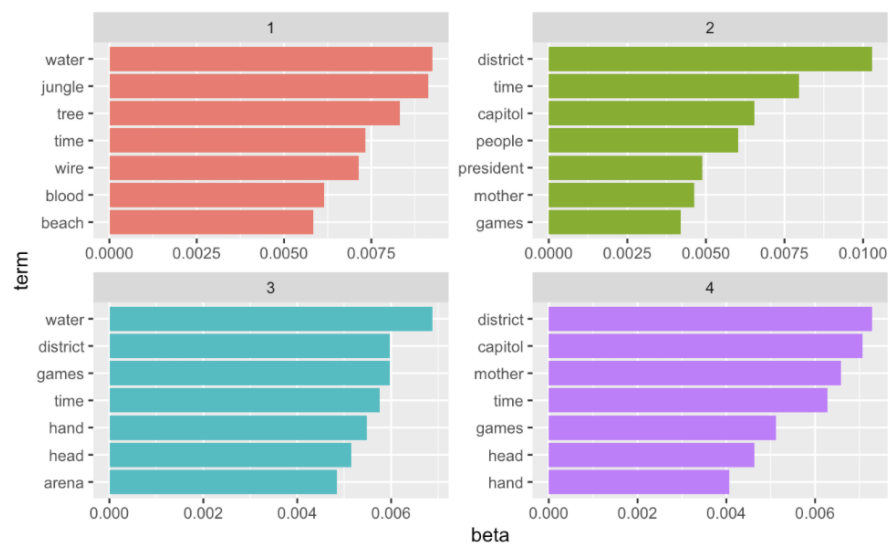


Figure 8: Themes Detected via LDA Topic Modeling in Book 2

The figure above illustrates the Book 2 themes identified via LDA modeling. These clusters were identified as: The Quarter Quell Arena (Topic 1), Political Unrest (Topic 2), Arena Combat (Topic 3), and Personal & Political Ties (Topic 4), representing the primary narrative pillars of the second installment.

3.3 Book 3: *Mockingjay*

3.3.1 Sentiment Over Time

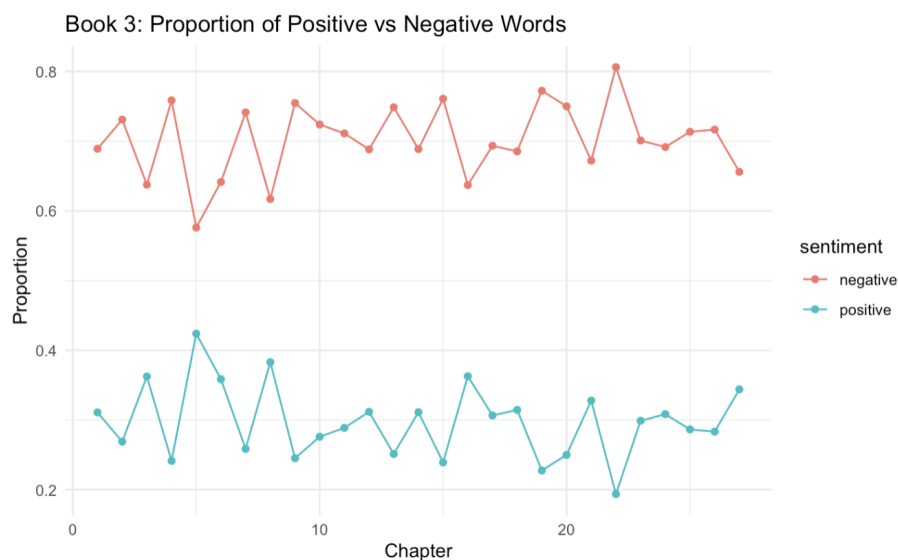


Figure 9: Proportion of Positive vs Negative Words in Book 3

The sentiment analysis for Book 3 reveals a stark and consistent predominance of negative sentiment, maintaining a high proportion of between 60% and 80% throughout the narrative. Unlike the fluctuating trends of the previous books, the negative skew in *Mockingjay* remains largely sustained across all chapters, reflecting the grim atmosphere of the series' conclusion.

3.3.2 NRC Sentiment

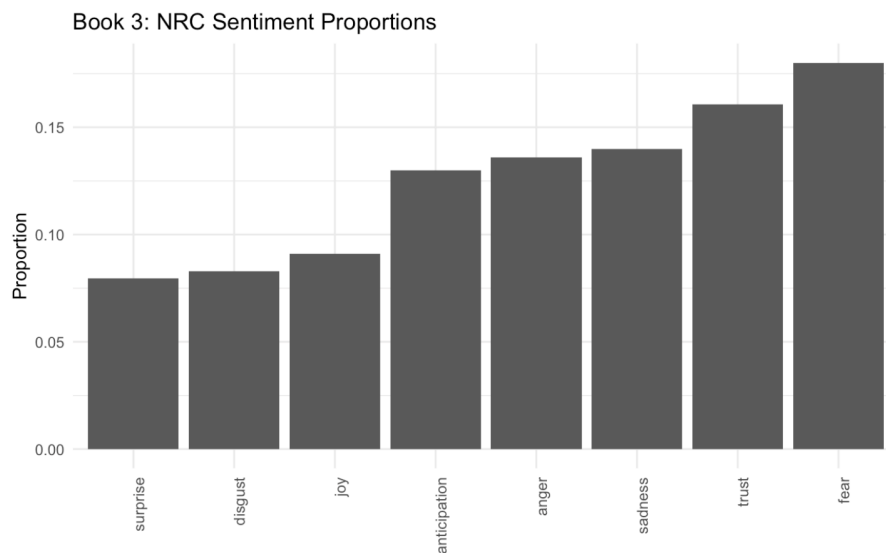


Figure 10: NRC Sentiment Analysis Book 3

The NRC sentiment analysis reveals that fear, trust, and sadness are the dominant emotional sentiments in Book 3 while surprise represents the least prevalent sentiment, suggesting that the narrative is driven more by sustained tension than by sudden plot shifts.

3.3.3 Vocabulary

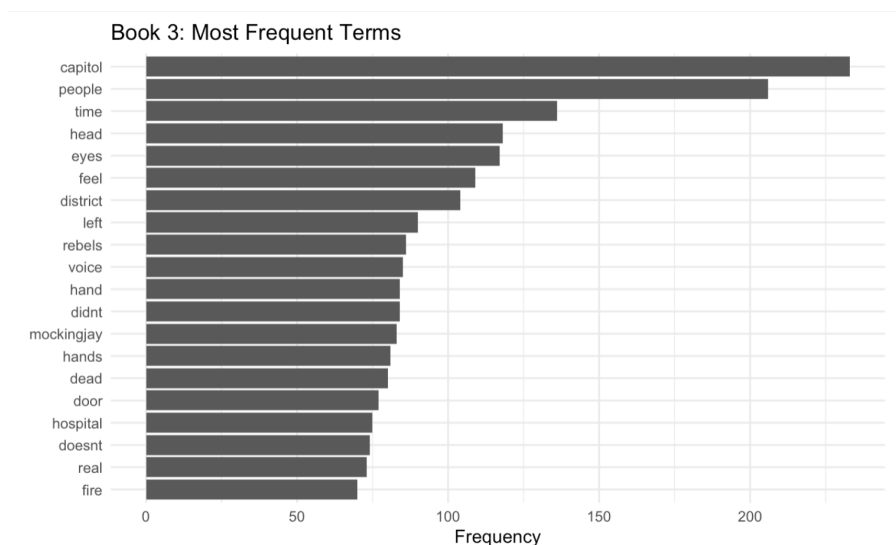


Figure 11: 20 Most Frequent Terms in Book 3

The frequency distribution for Book 3 highlights a significant thematic shift toward the broader political conflict and rebellion effort, with terms such as 'capitol,' 'rebels,' and 'people' emerging as dominant motifs.

3.3.4 Themes Detected Via Topic Modeling

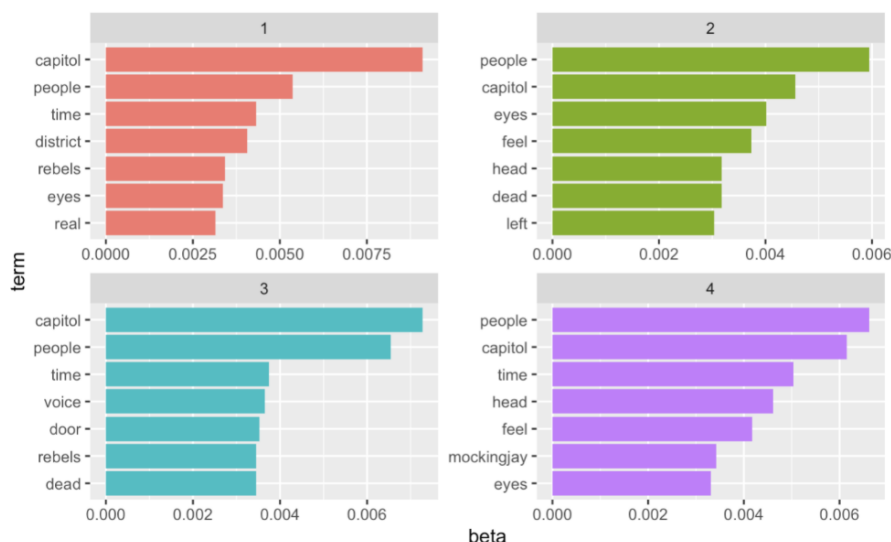


Figure 12: Themes Detected via LDA Topic Modeling in Book 3

The thematic landscape of the final book reflects the series' transition from survival and the arena to total warfare. The themes were classified as: The Rebellion & War Effort (Topic 1), Psychological Trauma (Topic 2), Assault on the Capital (Topic 3), Symbolism of the Mockingjay (Topic 4).

4 Cross-Book Comparisons

4.1 Summary Statistics

Book	Total Words	Unique Words	Lexical Diversity (%)
Hunger Games	99 893	7544	7.55
Catching Fire	101 719	7807	7.68
Mockingjay	101 458	8634	8.51

Table 1: Summary statistics across the trilogy

These statistics show that each book has around the same word count (about 100,000). It is apparent that Mockingjay has the most unique words and, consequently, the highest lexical diversity. This indicates a more varied vocabulary in the third book than in the other two, suggesting that the storyline is less repetitive and more complex. We will further explore if this change is correlated with differences in thematic analysis.

4.2 Writing Style Shifts Across the Trilogy

4.2.1 Chapter Lengths

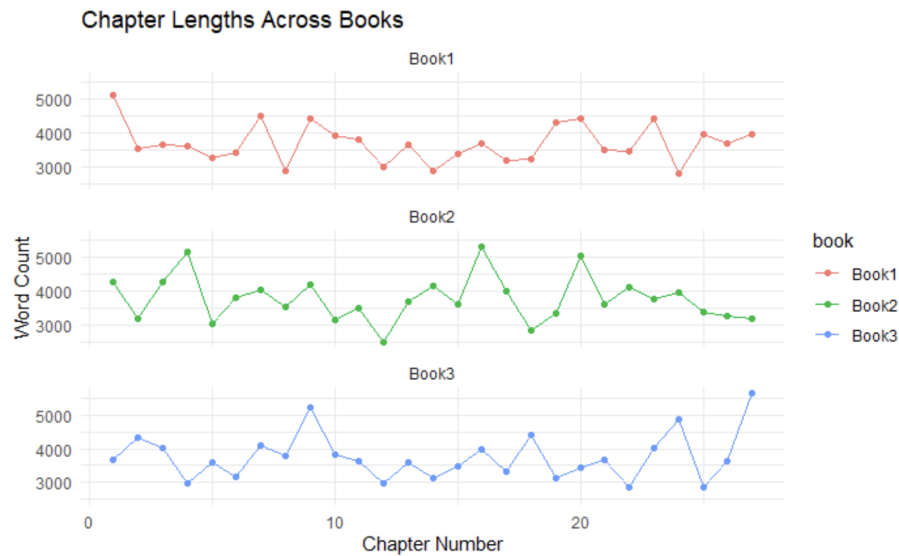


Figure 13: Differences in Chapter Lengths Across Trilogy

As the graphs show, chapter lengths are fairly consistent within each book (most fall between 3000 and 4500 words, with a few outliers). We also observe that each book exhibits substantial variability, with chapter lengths fluctuating, which may indicate a shift in narrative intensity.

4.2.2 Sentiment Overview

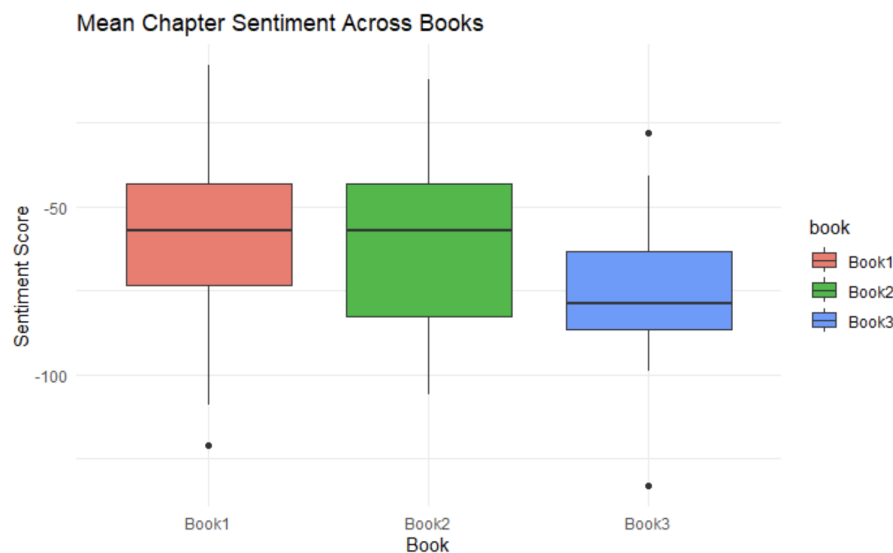


Figure 14: Chapter Sentiment Distribution Across Books

Using BING sentiment scores, the graph demonstrates a gradual decrease in sentiment across the trilogy, although none of the books appear to be positive. The median between

the first two books is quite similar, indicating a similar amount of narrative peril, but the second book does have a slight edge on the first, as the second quartile has a larger range of negativity. The third book features a significant drop in sentiment from the previous two, which might be attributed to raised stakes or intensity surrounding war and rebellion.

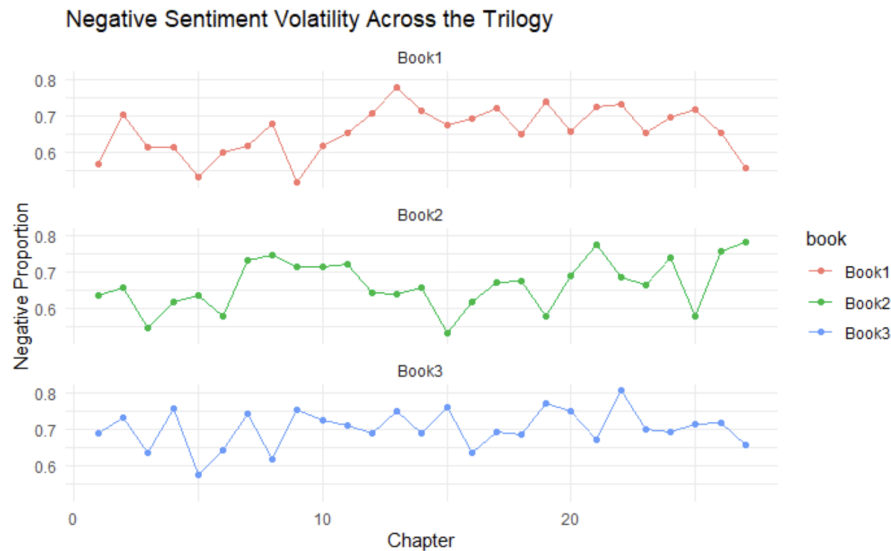


Figure 15: Sentiment Volatility Across Books

This graph illustrates that the amount of negative sentiment varies greatly within each book. In the first book, we see a stark increase about a third into it, which is likely the beginning of the game of death that the protagonist finds herself within. In the second book, this pattern is repeated, but closer to the beginning of it, which is likely when that book's respective game begins. The third book has the highest variability, with intensity going up and down chapter after chapter, which highlights how conflict is followed by a period of peace, and then repeating this rhythm.

4.2.3 Pronoun Shifts

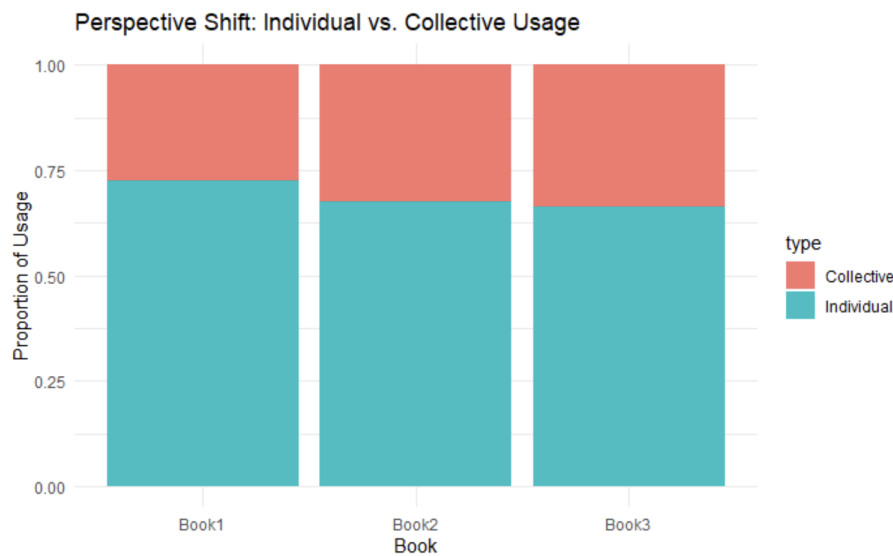


Figure 16: Proportions of Individual vs Collective Language

Over the course of the complete storyline, the narrative shifts from an individual's struggle for survival to a united group's battle for freedom. We want to see if this was reflected by the frequency of personal pronouns vs. collective, and this graph does show that the books become slightly more about the "we" and less about the "I." Still, every story is centered around one character, or told from their perspective, as personal pronouns appear more than 60 percent of the time in each.

4.2.4 Thematic Shifts

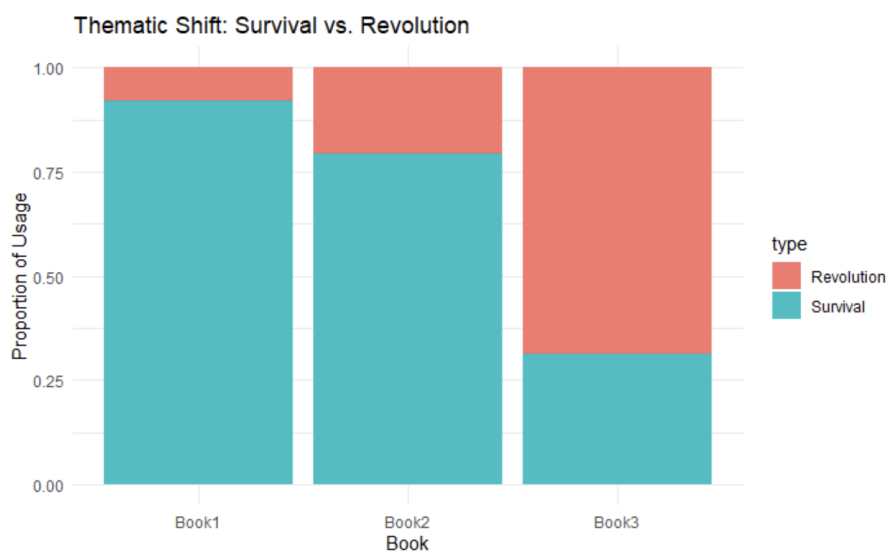


Figure 17: Proportion of Themes of Survival vs Revolution

Using some keywords for two themes, Survival and Revolution, we track how the narrative shifts from the first book to the third. As we can see, the books become less about survival,

the main theme of the first book, and more about revolution, the central premise of the third book. This is exactly what we would expect in our text analysis, as it is very apparent to a fan of the trilogy that this is the case.

4.2.5 Vocabulary Shifts

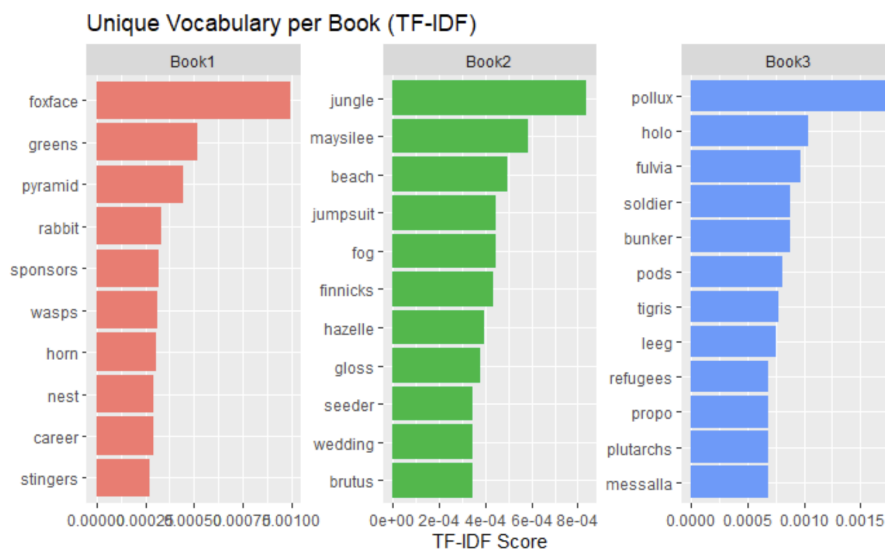


Figure 18: TF-IDF Scores Across Books

The unique words per book make a lot of sense, especially if you have read the books or seen the movies. The first book has words unique to the first game, like "rabbit", "wasps", and "Foxface" (a character who does not survive for the other two). This is clear in the second book as well, which takes place in a "jungle" with a "beach." The third book is where themes of war and revolution appear heavily, reflected by "soldier", "bunker", and "refugees."

4.3 Language Surrounding Main Characters

4.3.1 Bi-Gram Analysis

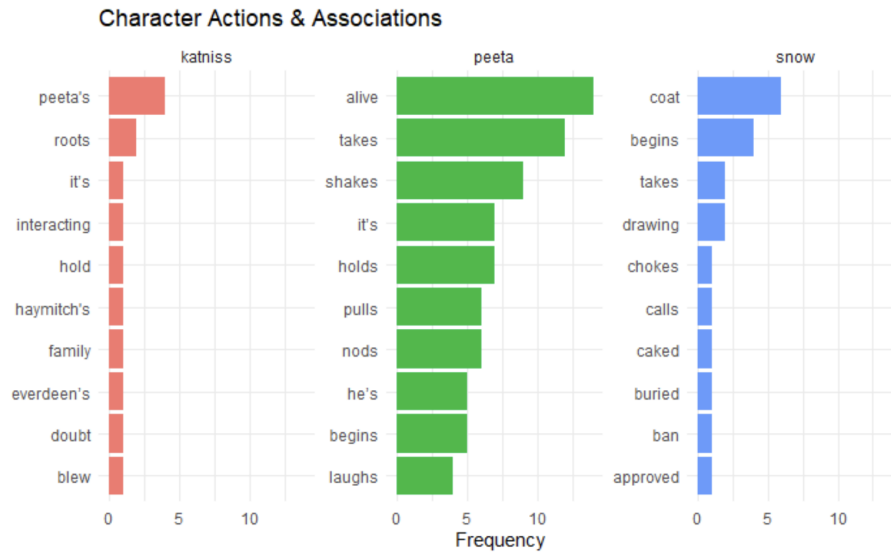


Figure 19: Bi-Gram Analysis

The bi-gram analysis highlights how each character is associated with different actions and contexts throughout the story. Katniss is often linked to interactions and relationships (e.g., “peeta’s,” “haymitch’s,” “family”), reinforcing her role as the central connector between characters. Peeta, on the other hand, is tied to more expressive and dynamic actions like “takes,” “shakes,” and “laughs,” which reflects his emotional and reactive nature. Snow’s associations are more cold and controlled (e.g., “coat,” “begins,” “chokes”), aligning with his authoritative and calculated presence in the narrative.

4.3.2 Sentiment Surrounding Main Characters

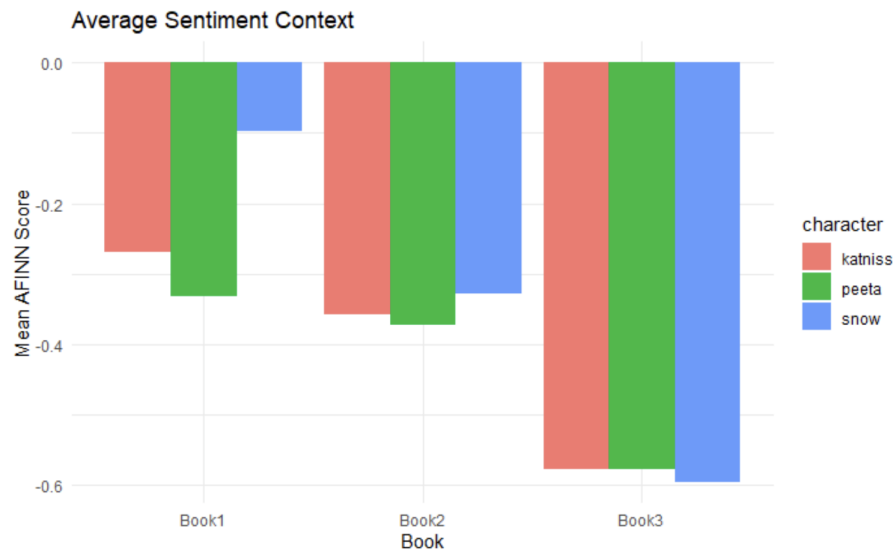


Figure 20: Average Sentiment Surrounding Each Main Character

The sentiment surrounding each character is negative across all three books, with scores becoming more negative over time. This reflects the darker tone of the series as it progresses, especially in the third book. By Mockingjay, all three characters are associated with similarly low sentiment, highlighting the intensity of the war-driven narrative.

4.4 Topics Across Books

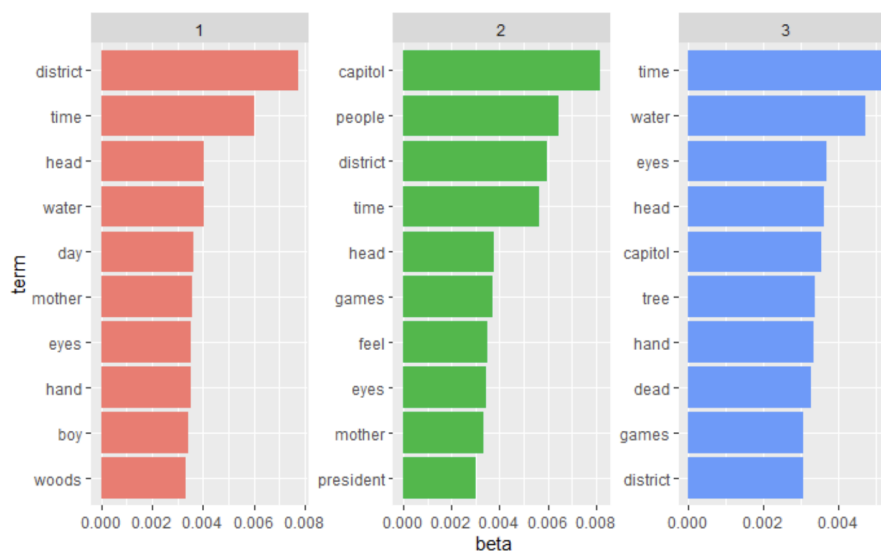


Figure 21: LDA of Topics in All Books

Topic 1 appears to focus on everyday life in the districts, with words like “district,” “mother,” and “day” suggesting a more personal and grounded setting. Topic 2 is centered around the Capitol and societal structure, highlighted by terms such as “capitol,” “people,” and “president.” Topic 3 seems more action- and survival-oriented, with words like “water,” “dead,” and “games” reflecting conflict and intensity.

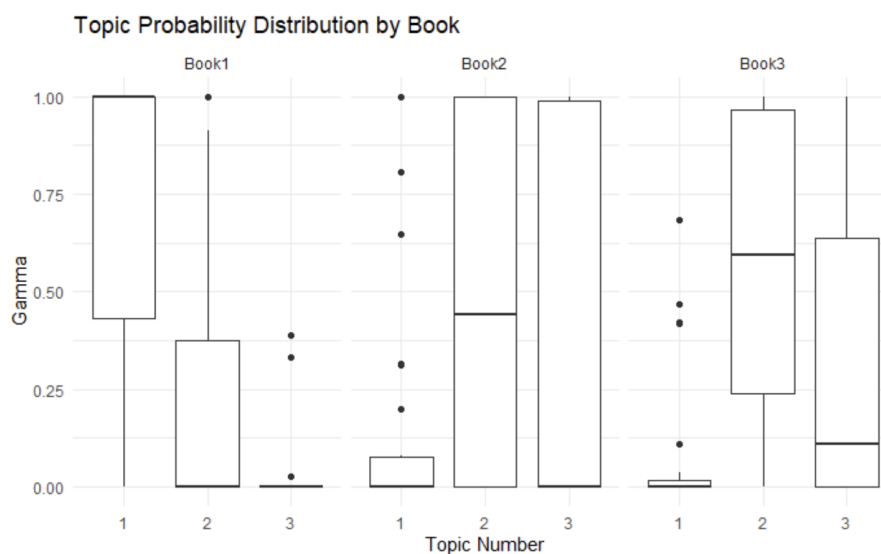


Figure 22: Distribution of Topics Across Books

The distribution shows that Topic 1 is most dominant in the first book, while Topics 2 and 3 appear more frequently in later books. In Book 2, the probabilities are more spread out, suggesting a mix of themes throughout the narrative. By Book 3, Topic 2 becomes the most prominent, reflecting a shift in focus as the story progresses.

4.5 Comparing Author to Herself

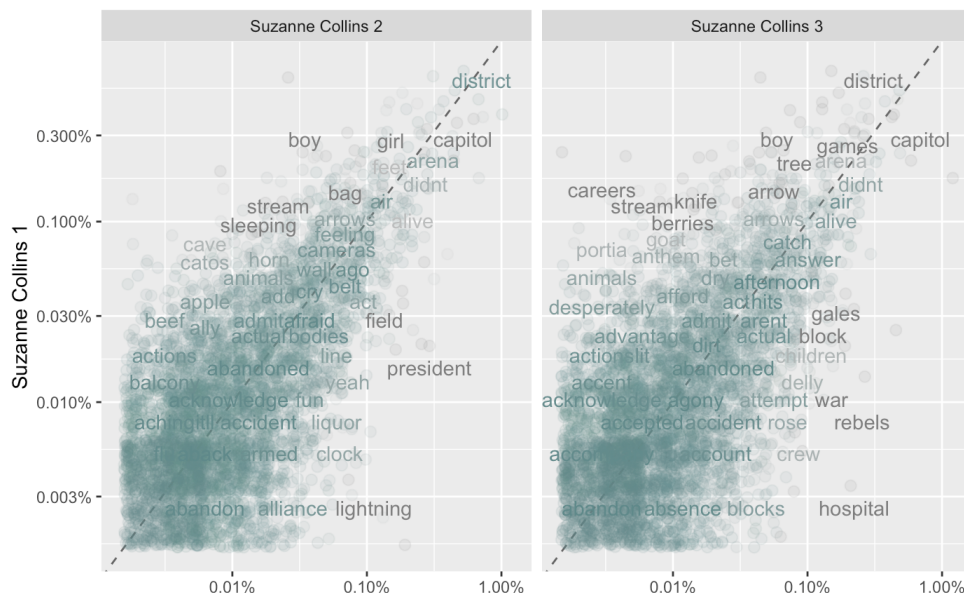


Figure 23: Scatterplot Comparison of Author to Herself

The scatterplots above, which compare the author's lexical choices across the trilogy, reveal a set of overlapping terms in all 3 books such as 'district,' 'capitol,' and 'arena.' Terms that are more distinct or prevalent to each book are also identified: 'careers' and 'stream' in Book 1 represent the survivalist focus of the initial Games; 'lightning' and 'alliance' in Book 2 signal at the Quarter Quell Arena, and 'rebels' and 'hospital' in Book 3 capture the narrative focus of war and rebellion.

4.6 Pairwise Comparison in Terms of Correlations

Comparison	cor	t	df	p-value	95% CI
Book 1 vs. Book 2	0.874	118.34	4328	< 0.001	[0.867, 0.881]
Book 1 vs. Book 3	0.749	74.50	4345	< 0.001	[0.736, 0.762]
Book 2 vs. Book 3	0.819	95.91	4515	< 0.001	[0.809, 0.828]

Table 2: Pearson Correlation Analysis of Word Frequencies Across Suzanne Collins' Books

Table 2 presents a pairwise correlation analysis of word frequencies across the trilogy. While all pairings demonstrate a strong positive correlation ($r > 0.70$, $p < 0.001$), the highest degree of lexical similarity exists between Book 1 and Book 2 ($r = 0.874$). In

contrast, the correlation weakens between Book 1 and Book 3 ($r = 0.749$), suggesting that the final installment introduces a more distinct vocabulary as the narrative shifts from the Games toward open warfare.

4.7 Classifications using Random Forest Modeling

Parameter	Value
Model Engine	ranger (Random Forest)
Number of Trees	500
Feature Extraction	TF-IDF (2,000 tokens)
Preprocessing	Tokenization, Stemming, Remove Stopwords & Non-Letters
Passage Length	100 words
Data Split	80% Training / 20% Testing

Table 3: Random Forest Classification Model Specifications

Metric	Result
Training Accuracy	1.000
Testing Accuracy	0.733

Phrase/Passage	Predicted Class
“I volunteer as tribute.”	Book 1 (The Hunger Games)
“I don’t want you forgetting how different our circumstances are... You’re my whole life,” Peeta says.	Book 2 (Catching Fire)
“Fire is catching! And if we burn, you burn with us!”	Book 1
“It’s impossible to be the Mockingjay... Snow will ensure that his life is much more worse than death.”	Book 3 (Mockingjay)

Table 4: Model Performance and Validation

The two tables above detail the classification task, which was performed using a Random Forest model trained on 100-word passages with an 80/20 data split. Table 3 outlines the specifications for the Random Forest classifier, utilizing a ranger engine with 500 trees and a TF-IDF vector space of 2,000 tokens. Although the model achieved a perfect training accuracy (suggesting a degree of overfitting with only 81 data points), the testing accuracy of 73.3% indicates a moderate ability to generalize across the trilogy. As shown in the validation samples (Table 4), the model correctly classifies key passages, though it incorrectly identified the iconic ‘Fire is catching!’ quote found in Book 3, suggesting substantial lexical overlap between the books.

5 Comparison to LLM

To compare large language model outputs with traditional text analysis, GPT-4 was provided with the top 75 most frequently used words for each book and tasked with identifying key words unique to each novel. The model produced a brief thematic analysis of the words it identified, highlighting the narrative focus of each book. Comparison with topic modeling results (LDA) revealed only minor overlap, suggesting that GPT-4 captured different aspects of word importance than the statistical methods. For example, the word “District” emerged as a key term in Book 3, corresponding to a central element of Topic 1 identified through LDA. However, relatively few words identified by TF-IDF were reflected in the GPT-4 output: none in Book 1, “Jungle” in Book 2, and “Pollux” in Book 3. These findings suggest that while large language models can offer useful thematic insights, they may emphasize different vocabulary patterns than traditional computational text mining techniques, highlighting the complementary nature of these approaches for literary analysis.

Here is the full LLM output: "Book 1: tributes, boy, girl, woods, arena, fire, bread, careers, gamemakers, stream. These words reflect the introduction of the concept of the Hunger Games and set a strong base for the dystopian world in which the story unfolds. Words like 'boy', 'girl', 'careers', 'tributes', and 'gamemakers' represent the main theme of children being selected to participate in deadly games, while 'woods', 'bread', 'fire', 'arena', and 'stream' help to construct the brutal, survivalist environment of the games.

Book 2: victors, jungle, door, family, house, lot, square, arm, skin, wire. These terms show a shift in the scope and scale of the story. Words like 'victors', 'jungle', 'door', 'house', 'square', and 'lot' suggest a wider social and geographical context, indicating a broader rebellion and a return to the deadly arena, this time with a unique 'jungle' environment. Furthermore, words like 'family', 'arm', 'skin', and 'wire' may reflect the growing resistance movement and a more personal, familial struggle.

Book 3: rebels, mockingjay, hospital, real, gales, pollux, war, peacekeepers, districts, hunger. These words, specific to the context of war and revolution, tell us about the climax of the story where the oppressed challenge their oppressors. 'Rebels', 'mockingjay', and 'war' explicitly discuss the rebellion and war, while 'hospital', 'real', 'gales', 'pollux', 'peacekeepers', and 'districts' provide insight into the broader society and the impact of the war. The word 'hunger', the premise upon which the series is built, makes its appearance in the last book, denoting the culmination of the struggle."

6 Statistical Analysis

To statistically evaluate differences across the trilogy, several statistical tests were conducted, including ANOVA, chi-square, and regression analyses. These tests were designed to quantify changes in sentiment and word usage across books and chapters.

6.1 ANOVA

Analysis of variance (ANOVA) was used to test whether the mean chapter sentiment differed significantly between the three books. Using the BING sentiment lexicon, we modeled sentiment score as a function of book:

$$\text{sentiment score} \sim \text{book}$$

Table 1 presents the ANOVA results. The test indicates that average chapter sentiment differs significantly across the books ($F(2, 78) = 3.956$, $p = 0.0231$), suggesting that the emotional tone evolves meaningfully as the narrative progresses.

	DF	Sum Sq	Mean Sq	F Value	Pr(>F)
Book	2	4958	2479.2	3.956	0.0231
Residuals	78	48 876	626.6		

Table 5: ANOVA test for chapter sentiment across books

6.2 Chi-Square Test

A chi-square test was applied to the top 50 most frequently used words to assess whether word distributions differed by book. The results ($\chi^2 = 744.48$, $df = 98$, $p < 2.2 \times 10^{-16}$) indicate statistically significant differences in vocabulary usage between the novels, reflecting shifts in thematic focus and narrative context.

χ^2	DF	p-value
744.48	98	$< 2.2e-16$

Table 6: Chi-square test for word distribution differences

6.3 Regression Analysis

Finally, a regression model was used to predict chapter sentiment based on book and chapter number:

$$\text{sentiment score} \sim \text{book} + \text{chapter}$$

The regression coefficients (Table 3) indicate that both book identity and chapter number contribute to variations in sentiment, with Book 3 showing a significant negative effect on sentiment compared to Book 1. The overall model statistics (Table 4) confirm that the model is significant ($F = 5.953$, $p = 0.00105$) and explains a meaningful portion of variance in sentiment across the trilogy.

Coefficient	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-44.1481	6.6072	-6.682	3.31×10^{-9}
bookBook2	-0.4444	6.4837	-0.069	0.945 53
bookBook3	-16.8148	6.4837	-2.593	0.011 37
chapter_number	-1.0265	0.3398	-3.020	0.003 42

Table 7: Regression coefficients predicting sentiment

Residual Std Error	Multiple R^2	Adjusted R^2	F-statistic	p-value
23.82	0.1883	0.1567	5.953	0.001 05

Table 8: Overall regression model statistics

Overall, these statistical analyses confirm that sentiment, word usage, and thematic emphasis differ significantly across the three books, providing quantitative support for the narrative and linguistic patterns observed in the trilogy.

7 Interpretation

The results of this analysis indicate a clear progression in tone, themes, and language across Suzanne Collins' *The Hunger Games* trilogy. Linguistic patterns and sentiment scores suggest that the emotional tone of the series becomes progressively darker as the narrative develops. While *The Hunger Games* focuses largely on immediate survival within the arena, *Catching Fire* introduces greater political tension and resistance, and *Mockingjay* shifts fully into themes of rebellion, warfare, and psychological trauma. The increasing negativity in sentiment scores, particularly in the third novel, reflects the escalating stakes and emotional weight of the narrative.

Character-level analysis also reveals distinct linguistic patterns associated with major characters. Word association results suggest that Katniss is most strongly linked with language related to relationships and loyalty, reflecting her role as the emotional center of the story. Peeta, by contrast, is more closely associated with language related to hope and compassion, reinforcing his symbolic role as a moral counterbalance to the violence of the Games. President Snow's language is strongly associated with conflict, power, and control, highlighting his position as the central antagonist in the trilogy.

The thematic progression is further supported by word frequency patterns and topic modeling results. The first two novels emphasize survival, arena combat, and interpersonal relationships, while the third novel places greater emphasis on revolution, military conflict, and the psychological consequences of war. These findings align with the narrative structure of the trilogy and demonstrate how computational text analysis can capture meaningful shifts in literary themes.

8 Limitations

Several limitations should be considered when interpreting the results of this study. First, preprocessing decisions, such as tokenization, stopword removal, and handling of punctuation, can influence word frequency counts and sentiment analysis outcomes. Different preprocessing choices might produce slightly different results, meaning that some patterns observed in this analysis may be partially dependent on these methodological decisions. Second, lexicon-based sentiment analysis, while useful for quantifying emotional tone, tends to oversimplify complex emotional nuance and context. Literary texts often convey subtle or mixed emotions that may not be fully captured by simple positive/negative scoring. Finally, while computational methods such as topic modeling, TF-IDF, and statistical tests provide valuable quantitative insights into language and thematic patterns, they cannot fully substitute for traditional literary interpretation. Certain aspects of narrative meaning, character development, and symbolic significance may be lost when reduced to statistical patterns. Future research could combine computational and close-reading approaches to provide a more holistic understanding of the text.

9 Conclusion

Overall, this study demonstrates that computational text mining methods can effectively identify linguistic and thematic developments across a literary series. Statistical testing, including ANOVA, chi-square tests, and regression analysis, confirmed that differences in sentiment, vocabulary, and thematic emphasis across the three novels are statistically significant. These results provide quantitative evidence for the narrative progression from individual survival to large-scale political rebellion within the trilogy.

Additionally, the TF-IDF analysis successfully identified distinctive vocabulary patterns across the novels, highlighting how certain terms become more prominent as the narrative evolves. In comparison with language generated by GPT-4, the TF-IDF results captured more unique and text-specific terms, suggesting that traditional statistical text mining methods remain highly effective for identifying distinctive linguistic patterns in literary analysis.

Taken together, these findings illustrate how computational approaches can complement traditional literary interpretation. By combining quantitative analysis with thematic interpretation, this project demonstrates how data-driven methods can reveal patterns in tone, character portrayal, and thematic development that may be difficult to observe through close reading alone.

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